



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

UNIT TEST

05-12-2021

JEE(Main + Advanced) : NURTURE COURSE [PHASE : S(TEPS)]

ANSWER KEY

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	B,C	A,B,C,D	A,C	A,B,C	B,C	B,C,D
SECTION-I (ii)	Q.	7	8	9	10		
	A.	A	A	A	C		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	2.0	1.33	0.61	55.36 to 55.42	0.09 to 0.10	0.4
SECTION-II (ii)	Q.	7	8	9			
	A.	4	25	1260			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,C,D	B	A,D	A,D	A,B,D	A,B,C
SECTION-I (ii)	Q.	7	8	9	10		
	A.	B	D	C	D		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	2.00	4.00	760.00	0.50	1.43	1.51
SECTION-II (ii)	Q.	7	8	9			
	A.	6	4	10			

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	B,C,D	B,C,D	A,B	A,B,C	A,B,C	A,B,D
SECTION-I (ii)	Q.	7	8	9	10		
	A.	C	D	A	B		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	10.00	8.65 or 8.66	2.00	2.82 or 2.83	6.00	16.00
SECTION-II (ii)	Q.	7	8	9			
	A.	8	5	772			

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

3. Ans (A,C)

$$qvB + k\ell = \frac{mv^2}{\ell}$$

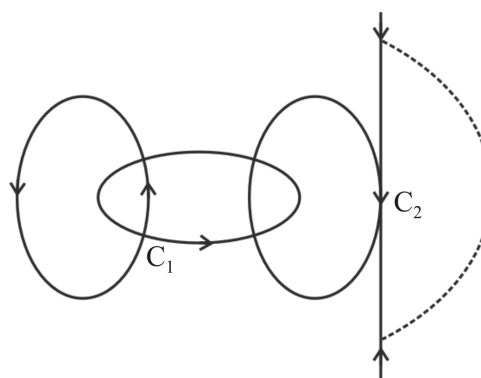
$$\Rightarrow v^2 - \frac{qB\ell}{m}v - \frac{k\ell^2}{m} = 0$$

$$v = \frac{\frac{qB\ell}{m} \pm \sqrt{\frac{q^2B^2\ell^2}{m^2} + \frac{4k\ell^2}{m}}}{2}$$

$$v = 15\hat{k} \text{ or } -10\hat{k}$$

4. Ans (A,B,C)

(B) & (D)



imaginary loop

$$\oint \vec{B} \cdot d\vec{\ell} = \mu_0 i_{en} = 0$$

$$\oint \vec{B} \cdot d\vec{\ell} = 0 \text{ semicircle as } B \propto \frac{1}{r^2}, \ell \propto r$$

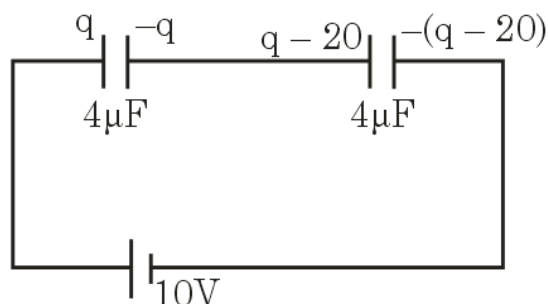
(C) at large distance, C_1 behaves like a dipole

for which 'B' is radially out. At nearby point, B

is down. So at same point, B is $\perp$.

$$(A) \oint \vec{B} \cdot d\vec{\ell} = \mu_0 i_{en} = 0$$

5. Ans (B,C)



$$\frac{q}{4} + \frac{1}{2}(q-20) = 10$$

$$q + 2q - 40 = 40\mu C$$

$$3q = 80\mu C \Rightarrow q = \frac{80}{3}\mu C$$

$$\text{Charge flow} = \frac{80}{3} - 20 = \frac{20}{3}\mu C$$

$$W_{\text{battery}} = \frac{20}{3} \times 10 = \frac{200}{3}\mu J$$

6. Ans (B,C,D)

$$I_1 = 3I_2 \Rightarrow I_2 = \frac{I}{4}, I_1 = \frac{3I}{4}$$

$$B = \frac{\mu_0 i}{4r} \text{ due to semicircular wire.}$$

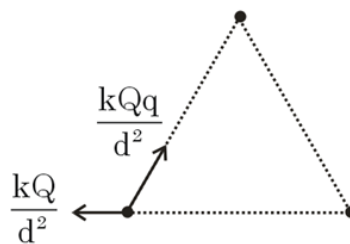
PART-1 : PHYSICS

SECTION-I (ii)

7. Ans (A)

$$F_r^2 = \left(\frac{k^2 Q}{d^2}\right)^2 + \left(\frac{kQq}{d^2}\right)^2 + 2\frac{k^2 Q^3 q}{d^4} \cos 120^\circ$$

$$= \frac{k^2 [Q^4 + Q^2 q^2 - Q^3 q]}{d^4}$$

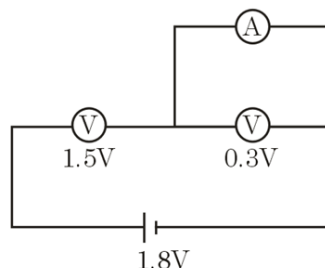


$$\frac{dF^2}{dq} = 0 \Rightarrow 2Q^2 q - Q^3 = 0$$

$$q = \frac{Q}{2}$$

9. Ans (A)

$$R_A = \frac{0.3}{0.5 \times 10^{-3}} = 600\Omega$$



$$0.3 = i_v \times R$$

$$1.5 = (i_v + 0.5 \times 10^{-3})R$$

$$1.5 = 0.3 + 0.5 \times 10^{-3} R$$

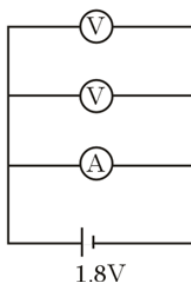
$$R = 2400\Omega$$

$$i = \frac{1.8}{2400 + 2400 + 600} = \frac{1.8}{5400} = \frac{1}{3000}$$

$$= 3.3 \times 10^{-4} A$$

$$v = 3.3 \times 10^{-4} \times 2400 = 0.8V$$

10. Ans (C)

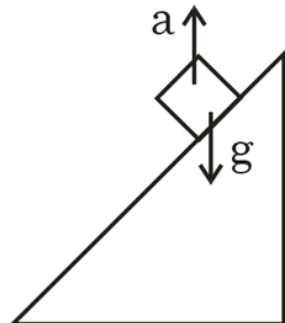


$$R_A = \frac{1.8}{600} = 3mA$$

PART-1 : PHYSICS

SECTION-II (i)

3. Ans (0.61)



$$\frac{1}{2} \times 16 \times \sin 30^\circ \times t^2 = 1.5$$

$$t = \frac{3}{8}; t = \frac{\sqrt{3}}{2\sqrt{2}} = \frac{1.73}{2.82} = 0.61 \text{ sec}$$

4. Ans (55.36 to 55.42)

$$N = m(g + a) \cos \theta$$

$$= 4 \times 16 \times \frac{\sqrt{3}}{2} = 32\sqrt{3} \text{ N}$$

$$55.36 \text{ to } 55.42 \text{ N}$$

5. Ans (0.09 to 0.10)

$$\frac{1}{2} m v_x^2 = qU_0$$

$$v_x = \sqrt{\frac{2qU_0}{m}}; t = \ell \sqrt{\frac{m}{2qU_0}}$$

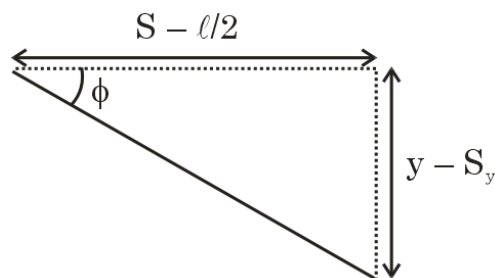
$$a_y = \frac{qU}{md}; v_y = a_y t = \frac{qU}{md} \times \ell \sqrt{\frac{m}{2qU_0}}$$

$$\tan \phi = \frac{v_y}{v_x} = \frac{\frac{qU}{md} \ell \sqrt{\frac{m}{2qU_0}}}{\sqrt{\frac{2qU_0}{m}}}$$

$$\tan \phi = \frac{qU}{md} \ell \times \frac{m}{2qU_0} = \frac{U\ell}{2dU_0} = \frac{50 \times 2}{2 \times 0.5 \times 1000}$$

6. Ans (0.4)

$$S_y = \frac{1}{2} a_y t^2 = \frac{qU}{2md} \times \frac{\ell^2 m}{2qU_0} = \frac{U\ell^2}{4dU_0}$$



$$\frac{y - S_y}{S - \frac{\ell}{2}} = \tan \phi$$

$$y - \frac{U\ell^2}{4dU_0} = \left(S - \frac{\ell}{2} \right) \times \frac{U\ell}{2dU_0}$$

$$y = \left[\left(S - \frac{\ell}{2} \right) \times \frac{\ell}{2d} + \frac{\ell^2}{4d} \right] \frac{U}{U_0}$$

$$= \left(\frac{S\ell}{2d} \right) \frac{U}{U_0}$$

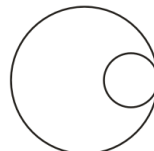
$$\frac{dy}{dU} = \frac{S\ell}{2dU_0} = \frac{0.2 \times 0.02}{2 \times 0.5 \times 10^{-2} \times 1000} \text{ m/V}$$

$$= 0.4 \text{ mm/V}$$

PART-1 : PHYSICS

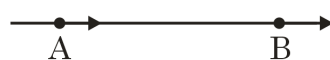
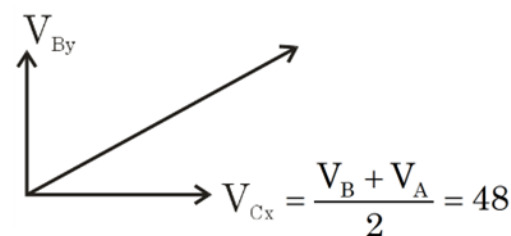
SECTION-II (ii)

7. Ans (4)



$$E = \frac{\rho R}{2\epsilon_0} - 0 = \frac{\rho R}{4\epsilon_0}$$

8. Ans (25)



$$V_{Cy}^2 + 48^2 = 52^2$$

$$V_{Cy} = 20 \text{ mph}$$

$$V_{CBx} = 63 - 48 = 15 \text{ mph}$$

$$V_{Cy} = V_{CBx} = 20 \text{ mph}$$

$$V_{CB} = \sqrt{V_{CBx}^2 + V_{CBx}^2} = 25 \text{ mph}$$

9. Ans (1260)

Potential difference without introducing the slab is given by

$$V_0 = E \times d = 300 \times 5 = 1500 \text{ volt}$$

The capacitance C_0 is given by

$$C_0 = \frac{\epsilon_0 A}{d} = \frac{\epsilon_0 A}{0.05} \text{ farad}$$

$$\therefore \epsilon_0 A = 0.05 C_0 \text{ ..(1)}$$

when the slab is inserted, the capacitance C is given by

$$C = \frac{\epsilon_0 A}{d - t + \frac{t}{K}} \text{ where } t = \text{thickness of slab}$$

$$= \frac{0.05 C_0}{0.04 + \frac{0.01}{5}} = \frac{25 C_0}{21} \text{(2)}$$

$$\text{Now } C_0 V_0 = CV$$

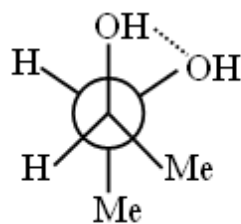
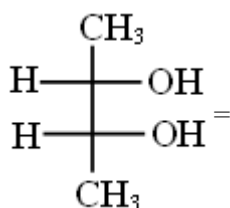
$$C_0 \times 1500 = \frac{25 C_0}{21} \times V$$

$$\text{or } V = \frac{1500 \times 21}{25} = 1260 \text{ volt}$$

PART-2 : CHEMISTRY

SECTION-I (i)

2. Ans (B)



Most stable
conformation
with H-bonding

Meso butane -2,3,-diol

3. Ans (A,D)

$$(A) \frac{C_B}{C_C} = \frac{k_1}{k_2} = \frac{1}{2} \text{(1)}$$

$$\text{Also } C_A = C_{A_0} - C_B - C_C$$

$$\text{or, } 0 = 2 - C_B - C_C$$

$$\text{or, } C_B + C_C = 2 \text{(2)}$$

$$\text{or, } C_B + 2C_B = 2$$

$$\text{or, } C_B = \frac{2}{3}$$

$$\therefore C_C = \frac{4}{3}$$

$$\therefore \theta = \frac{2}{3} \times (-72) + \frac{4}{3} \times 42$$

$$= -48 + 56 = 8^\circ \Rightarrow \text{dextrorotatory} \Rightarrow A \text{ is}$$

correct

(B) Wrong

$$(C) t_{1/2} = \frac{\ln 2}{(k_1 + k_2)} = \frac{0.693}{6.93 \times 10^{-2} + 13.86 \times 10^{-2}} = \frac{10}{3} \text{ min.}$$

$\therefore$ (C) is wrong

$$(D) C_A = C_{A_0} - C_B - C_C$$

$$\text{or, } 0.5 = 2 - C_B - C_C$$

$$\Rightarrow C_B + C_C = 1.5$$

$$\text{or } C_B + 2C_B = 1.5$$

$$\text{or, } C_B = 0.5$$

$$\therefore C_C = 1$$

$$\therefore \theta = 0.5 \times 60 + 0.5 (-72) + 1 \times 42$$

$$= 30 - 36 + 42 = 36^\circ$$

4. Ans (A,D)

Process 1st

$$PV^{3/2} = K_1$$

$$\Rightarrow TV^{x-1} = \text{const.}, x = 3/2$$

$$\Rightarrow T_1 V_1^{1/2} = T_2 V_2^{1/2}$$

$$\text{or, } T_2 = T_1 \left(\frac{V_1}{V_2} \right)^{1/2} = T_1 \left(\frac{V_1}{2V_1} \right)^{1/2}$$

$$= \sqrt{0.5} T_1, 0.71 T_1$$

2nd Process

$$x = 2$$

$$T_1 V_1 = T_2 V_2 = T_2 = T_1 \left(\frac{V_1}{V_2} \right) = T_1 \left(\frac{1}{2} \right) = 0.5 T_1$$

Clearly, T_2 1st process > T_2 2nd process

$$\therefore |\Delta T| \text{ 2nd process} > |\Delta T| \text{ 1st process}$$

$$\therefore |\Delta U| \text{ 2nd process} > |\Delta U| \text{ 1st process}$$

(B) 1st process

$$P_1 V_1^{3/2} = P_2 V_2^{3/2}$$

$$\text{or, } P_2 = P_1 \left(\frac{V_1}{V_2} \right)^{3/2} = P_1 \left(\frac{V_1}{2V_1} \right)^{3/2} = \frac{P_1}{2\sqrt{2}}$$

$$\text{2nd process, } P_1 V_1^2 = P_2 V_2^2$$

$$\text{or, } P_2 = P_1 \left(\frac{V_1}{2V_1} \right)^2 = \frac{P_1}{4}$$

Clearly (P_2) 1st process > (P_2) 2nd process

$$(C) (C_m)_{1st} = \frac{3}{2}R + \frac{R}{1-3/2} = \frac{3}{2}R - 2R = -0.5R$$

$$(C_m)_{2nd} \text{ process} = \frac{3}{2}R + \frac{R}{1-2} = 0.5R$$

$\therefore \Delta T$ is -ve in both $\therefore$ heat is absorbed in 1st process and it is released in 2nd process.

$$(D) w_{1st \text{ process}} = \frac{nR\Delta T}{(x-1)} = \frac{nR\Delta T}{(3/2-1)}$$

$$= 2nR\Delta T = 2nR(0.71 T_1 - T_1) = -0.58 nRT_1$$

$$w_{2nd \text{ process}} = \frac{nR\Delta T}{(x-1)} = \frac{nR\Delta T}{(2-1)} = nR\Delta T =$$

$$nR(0.5 T_1 - T_1) = -0.5 nRT_1$$

Clearly $|w_{1st \text{ process}}| > |w_{2nd \text{ process}}|$

PART-2 : CHEMISTRY

SECTION-I (ii)

7. Ans (B)

$$\therefore E_{n_2} > E_{n_1} \Rightarrow n_2 > n_1$$

$$\frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2} = 6 \text{ \&}$$

$$\frac{n_2(n_2 - 1)}{2} = 15$$

$$\Rightarrow n_2 = 6 \text{ \& } n_1 = 3.$$

8. Ans (D)

$$n_2 \sum (n_2 - n_1) = 6$$

$$\frac{n_2(n_2 - 1)}{2} = 15 \Rightarrow n_2 = 6$$

$$n_1 \sum (6 - n_1) = 6 \Rightarrow n_1 = 3$$

$$\Delta E = 13.6 z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

$$\text{or, } 4.53 = 13.6 z^2 \left[\frac{1}{9} - \frac{1}{36} \right]$$

$$\text{or, } 4.53 = 13.6 z^2 \times \frac{4-1}{36} = \frac{13.6 z^2}{12}$$

$$\text{or, } z^2 = \frac{4.53 \times 12}{13.6} = 4 \Rightarrow z = 2$$

For minimum wavelength :

$$n_2 = 6, n_1 = 1$$

$$\frac{1}{\lambda} = Rz^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

$$= R \times 4 \left[1 - \frac{1}{36} \right]$$

$$= \frac{4R \times 35}{36} = \frac{35R}{9}$$

$$\text{or, } \lambda = \frac{9}{35R}$$

PART-2 : CHEMISTRY

SECTION-II (i)

3. **Ans (760.00)**

Let x moles of NH_3 is produced

$$\therefore x \times 1 = 0.5 \times \frac{2}{3} \times 3 \Rightarrow x = 1 \text{ mole}$$

$$\therefore \text{CO}_2 = \frac{1}{2} \text{ mole}$$

$$P = \frac{nRT}{V} = \frac{1.5 \times 0.08 \times 300}{36} = 1 \text{ atm}$$

$$\therefore P_{\text{total}} = P_{\text{initial}} + P_{\text{NH}_3 + \text{CO}_2} = 1 + 1 = 2 \text{ atm}$$

$$\therefore \Delta P = 1 \text{ atm} \Rightarrow 760 \text{ mm}$$

PART-2 : CHEMISTRY

SECTION-II (ii)

7. **Ans (6)**

(iii), (iv), (v), (vi), (vii), (viii)

9. **Ans (10)**

$$x = 2$$

$$y = 8$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (B,C,D)**

$$(A) f(x) = \begin{cases} e^x - e^{-x} + 2 & x \geq 0 \\ 2 & x < 0 \end{cases}$$

$\therefore$ Non injective

$$(B) f(x) = \frac{2^x + 1}{(2^x)^2 - 1} = \frac{1}{2^x - 1} \text{ Injective}$$

$$(C) f(x) = 4x - |\cos x|$$

$f'(x)$ is positive $\therefore$ injective

$$(D) f(x) = x^3 - 2x^2 + 5x + 6$$

$$\Rightarrow f'(x) = 3x^2 - 4x + 5 \text{ which is always}$$

positive $\therefore$ Injective

2. **Ans (B,C,D)**

$$\frac{x^2}{25} + \frac{y^2}{24} = 1, a = 5, b = 2\sqrt{6}, e = \frac{1}{5}$$

So, the point is on the directrix $x = \frac{a}{e} = 25$.

So AB is a focal chord, also as

$$\frac{1}{SA} + \frac{1}{SB} = \frac{2a}{b^2}$$

$$\Rightarrow \text{if } S_1A = \frac{60}{13} \text{ then } S_1B = 5.$$

$$\text{Also } S_1A + S_2A = 2a = 10 = S_1B + S_2B$$

$$\Rightarrow S_2A = \frac{70}{13}, S_2B = 5. \text{ Distance from directrix}$$

$$= \frac{1}{e} \times \text{distance from focus} = \frac{70}{13} \times 5 = \frac{350}{13}$$

3. **Ans (A,B)**

The total score of 4 bottom players is atleast

${}^4C_2 = 6 \Rightarrow$ The score of player placing II is atleast 6 but at most 7. So three possibilities.

$$\begin{matrix} 6 & 6\frac{1}{2} & 7 \\ \downarrow & \text{(rulled out)} & \text{(rulled out)} \end{matrix}$$

player II gets if $6\frac{1}{2}$, then I if 7, player

6 points player also gets must be

$$6\frac{1}{2} \text{ (not possible) winner}$$

$\Rightarrow$ Total score of top four players is

$${}^8C_2 - 6 = 22$$

$\Rightarrow$ winner gets 7 points

Players Score Score

$$P_1 \quad 7 \quad 6.5$$

$$P_2 \quad 6 \quad 6$$

$$P_3 \quad 5 \quad 5.5$$

$$P_4 \quad 4 \quad 4$$

$$P_5 \quad 3 \quad 3$$

$$P_6 \quad 2 \quad 2$$

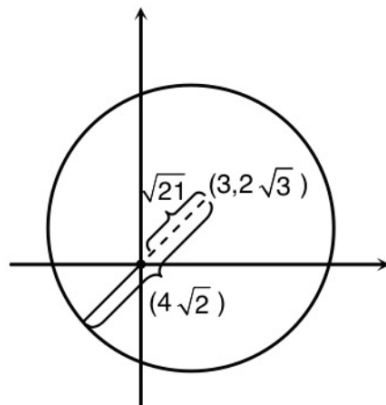
$$P_7 \quad 1 \quad 1$$

$$P_8 \quad 0 \quad 0$$

$\Rightarrow$ Score of third & fourth player is

$$22 - 7 - 6 = 9$$

4. Ans (A,B,C)



$$\frac{2b^2}{a} = 4, e^2 = 1 - \frac{b^2}{a^2} = \frac{1}{2}$$

$$\Rightarrow a = 4$$

$$2ae = 2.4 \frac{1}{\sqrt{2}} = 4\sqrt{2}$$

so other focus lies on circle

$$(x-3)^2 + (y-2\sqrt{3})^2 = (4\sqrt{2})^2$$

5. Ans (A,B,C)

Number beginning with 2 and not divisible by

$$5 \text{ is } (6! - 5!) = 600$$

Number beginning with 3 and not divisible by

$$5 \text{ is } (6! - 5!) = 600$$

Number beginning with 5

$$\text{is } 6! = 720$$

$$600 + 600 + 720 = 1920$$

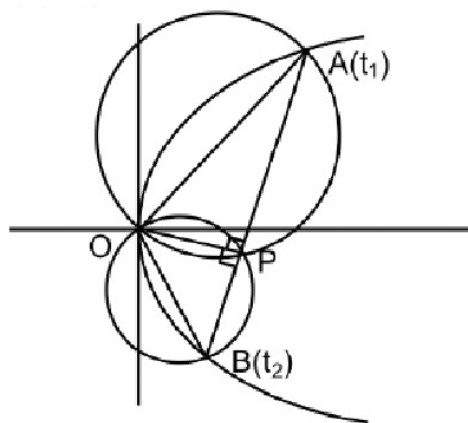
Number beginning with 6

$$\boxed{6} \boxed{2} \boxed{} \boxed{} \boxed{} \boxed{} \boxed{} = 5! - 4! = 96$$

$$\text{Total} = 2016$$

$$2017^{\text{th}} \text{ number} = 6325789$$

6. Ans (A,B,D)



Since OA & OB are diameters of circles

$$\angle OPA = \angle OPB = 90^\circ$$

Hence A, P, B are collinear

$$\text{Now } m = \frac{2}{t_1 + t_2} = \frac{2}{\frac{1}{m_1} + \frac{1}{m_2}} = \frac{2m_1 m_2}{m_1 + m_2}$$

$$\left(m_1 = \frac{1}{t_1} \& m_2 = \frac{1}{t_2} \right)$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (C)

$$T_r = \frac{(x+r+1) - (x+r)}{(x+r)(x+r+1)} = \frac{1}{x+r} - \frac{1}{x+r+1}$$

$$\sum_{r=1}^n T_r = \frac{1}{x+1} - \frac{1}{x+n+1} = \frac{n}{(x+1)(x+n+1)}$$

$$f(x) = \log_e \left(\frac{n}{(x+1)(x+n+1)} \right)$$

$$\frac{9}{(x+1)(x+10)} > 0 \Rightarrow x \in (-\infty, -10) \cup (-1, \infty)$$

8. Ans (D)

$$T_r = \frac{(x+r+1) - (x+r)}{(x+r)(x+r+1)} = \frac{1}{x+r} - \frac{1}{x+r+1}$$

$$\sum_{r=1}^n T_r = \frac{1}{x+1} - \frac{1}{x+n+1} = \frac{n}{(x+1)(x+n+1)}$$

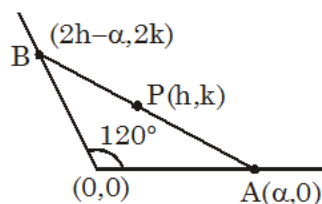
$$f(x) = \log_e \left(\frac{n}{(x+1)(x+n+1)} \right)$$

$\forall n \in \mathbb{N}$. Range is $\mathbb{R}$.

PART-3 : MATHEMATICS

SECTION-II (i)

1. Ans (10.00)



Let $(\alpha, 0) \Rightarrow B(2h - \alpha, 2k)$

$$\frac{2k}{2h - \alpha} = -\sqrt{3} \Rightarrow \alpha = \frac{2k + 2\sqrt{3}h}{\sqrt{3}}$$

$$4k^2 + (2h - 2\alpha)^2 = 100$$

$$3x^2 + 7y^2 + 4\sqrt{3}xy = 75$$

Let $x = r\cos\theta$, $y = r\sin\theta$

$$\Rightarrow r^2 = \frac{75}{3\cos^2\theta + 7\sin^2\theta + 4\sqrt{3}\sin\theta\cos\theta}$$

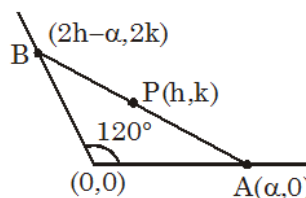
$$r^2 = \frac{75}{3 + 4\sin^2\theta + 2\sqrt{3}\sin 2\theta}$$

$$r^2 = \frac{75}{5 + (2\sqrt{3}\sin 2\theta - 2\cos 2\theta)}$$

$$r_{\max}^2 = \frac{75}{5 + (-4)} = 75$$

$$\Rightarrow r_{\max} = 5\sqrt{3}$$

2. Ans (8.65 or 8.66)



Let $(\alpha, 0) \Rightarrow B(2h - \alpha, 2k)$

$$\frac{2k}{2h - \alpha} = -\sqrt{3} \Rightarrow \alpha = \frac{2k + 2\sqrt{3}h}{\sqrt{3}}$$

$$4k^2 + (2h - 2\alpha)^2 = 100$$

$$3x^2 + 7y^2 + 4\sqrt{3}xy = 75$$

Let $x = r\cos\theta$, $y = r\sin\theta$

$$\Rightarrow r^2 = \frac{75}{3\cos^2\theta + 7\sin^2\theta + 4\sqrt{3}\sin\theta\cos\theta}$$

$$r^2 = \frac{75}{3 + 4\sin^2\theta + 2\sqrt{3}\sin 2\theta}$$

$$r^2 = \frac{75}{5 + (2\sqrt{3}\sin 2\theta - 2\cos 2\theta)}$$

$$r_{\max}^2 = \frac{75}{5 + (-4)} = 75$$

$$\Rightarrow r_{\max} = 5\sqrt{3}$$

9. Ans (A)

$$\sum_{r=0}^n (-1)^r \cdot a_r \cdot {}^nC_r = a_0 {}^nC_0 - a_1 {}^nC_1 + a_2 {}^nC_2 - \dots + (-1)^n a_n {}^nC_n$$

= coefficient of the constant term in

$$[a_0 + a_1x + a_2x^2 + \dots + a_{2n}x^{2n}] \times \left[{}^nC_0 - {}^nC_1\left(\frac{1}{x}\right) + {}^nC_2\left(\frac{1}{x}\right)^2 - \dots + (-1)^n {}^nC_n\left(\frac{1}{x}\right)^n \right]$$

= coefficient of the constant term in

$$(1 + x + x^2)^n \left(1 - \frac{1}{x}\right)^n$$

= coefficient of x^n in $(x^2 + x + 1)^n (x - 1)^n$

= coefficient of x^n in $(x^3 - 1)^n$

= 0 as n is not a multiple of 3.

10. Ans (B)

Dividing both the sides of (1) by x^n , we get

$$\left(1 + x + \frac{1}{x}\right)^n = \sum_{r=0}^{2n} a_r r^{r-n}$$

Thus,

a_n = constant term in the expansion of

$$\left(1 + x + \frac{1}{x}\right)^n$$

But

$$\left(1 + x + \frac{1}{x}\right)^n = C_0 + C_1 \left(x + \frac{1}{x}\right) + C_2$$

$$\left(x + \frac{1}{x}\right)^2 + \dots + C_n \left(x + \frac{1}{x}\right)^n$$

Thus,

$$a_n = C_0 + C_2({}^2C_1) + C_4({}^4C_2) + \dots \Rightarrow a_n = 1 + a$$

3. Ans (2.00)

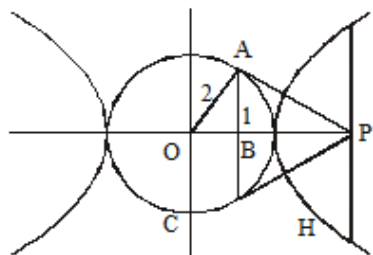
$$\frac{1}{e_1^2} + \frac{1}{e_2^2} = 1 \therefore e_1 = e_2$$

$$\Rightarrow e_1 = \sqrt{2} \Rightarrow a = b$$

foot of perpendicular drawn from focus upon any tangent always lies on auxiliary circle.

$$x^2 + y^2 = a^2 \Rightarrow a = 2 \Rightarrow b = 2$$

C : $x^2 + y^2 = 4$ (which is auxiliary circle itself)



Number of common tangent of H & C is 2

4. Ans (2.82 or 2.83)

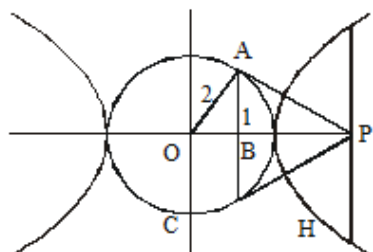
$$\frac{1}{e_1^2} + \frac{1}{e_2^2} = 1 \therefore e_1 = e_2$$

$$\Rightarrow e_1 = \sqrt{2} \Rightarrow a = b$$

foot of perpendicular drawn from focus upon any tangent always lies on auxiliary circle.

$$x^2 + y^2 = a^2 \Rightarrow a = 2 \Rightarrow b = 2$$

C : $x^2 + y^2 = 4$ (which is auxiliary circle itself)



If ellipse & hyperbola cuts orthogonally
 $\Rightarrow$ confocal.

$$\text{foci of H} \Rightarrow (\pm 2\sqrt{2}, 0)$$

$$\Rightarrow (4e_e, 0) = (2\sqrt{2}, 0)$$

$$e_e = \frac{1}{\sqrt{2}}$$

$$\beta^2 = 16 \left(1 - \frac{1}{2} \right) = 8 \Rightarrow \beta = 2\sqrt{2}$$

5. Ans (6.00)

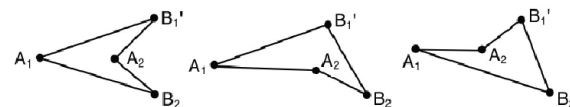
$$({}^{2n}C_2 \times {}^{2n}C_1)2 = 4n^2(2n-1)$$

$$\Rightarrow 8n^3 - 4n^2 = 1584$$

$$\Rightarrow 2n^3 - n^2 - 396 = 0 \Rightarrow n = 6$$

6. Ans (16.00)

$$\underbrace{{}^nC_2 \times {}^nC_2 \times 4 + {}^nC_1 \times {}^nC_1 \times {}^nC_1 \times {}^nC_1 + {}^nC_2 \times {}^nC_1 \times {}^nC_1 \times 4 \times 3}_{\text{Convex}} + \underbrace{{}^nC_2 \times {}^nC_1 \times {}^nC_1 \times 4 \times 3}_{\text{Concave}}$$



$$\Rightarrow 8n^4 - 8n^3 + n^2$$

$$\Rightarrow \lambda = 8 \text{ and } \beta = 8$$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. Ans (8)

$$3[x] - 4x = 4 \quad \text{or} \quad 3[x] - 4x = -4$$

$$\{x\} = \frac{-4 - [x]}{4} \quad \{x\} = \frac{4 - [x]}{4}$$

$$-8 < [x] \leq -4$$

$$0 < [x] \leq 4$$

$$[x] = -4, -5, -6, -7$$

$$[x] = 1, 2, 3, 4$$

$$\{x\} = 0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}$$

$$\{x\} = \frac{3}{4}, \frac{1}{2}, \frac{1}{4}, 0$$

4 solutions

4 solutions .

8. Ans (5)

$$10^6 - 1 = 999999 = 7 \times 142857$$

$$\text{If } N = 6K + r$$

$$10^N - 10^r = 10^{6k+r} - 10^r = 10^r(10^{6k} - 1)$$

$$= 10^r(10^6 - 1)(10^{6k-6} + 10^{6k-12} + \dots + 10^0 + 1)$$

Division of any integral power of 10 by 6 leaves a remainder equal to 4. Therefore

$$10^n - 4 = \underbrace{999 - 96}_{n \text{ times}} \text{ is always divisible by 6.}$$

$$10^n = 6k + 4$$

so

$$10^{10} + 10^{10^2} + 10^{10^3} + 10^{10^{10}} = 10^{6k_1+4} + 10^{6k_2+4} + \dots + 10^{6k_{10}+4}$$

where it is divided 7 it leaves

$$10^4 + 10^4 + \dots 10 \text{ times}$$

$$= 10 \times 10^4 = 10^5 = 10,000 = 7 \times 14285 + 5$$

we remained is 5.

9. Ans (772)

n(A) = number of elements in A = m

n(B) = number of elements in B = 12 - m

m $\neq$ 0, 6, 12

For A, we have 10 elements to choose from

$$N = \sum_{m=1}^{11} {}^{10}C_{m-1} - {}^{10}C_5 = 772$$